



Advanced Engineering Maths

Advanced Engineering Mathematics

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You have briefly met these in year 1 and our purpose here is to expand and explore the subject more fully.

We will assume virtually no knowledge from year 1

Where shall we start ?

Either, we immediately digress into a fairly lengthy discussion on practical problems that benefit from matrix approaches or we approach matrices as an abstract subject and then finish with applications.



Neither extreme is ideal, so we'll seek some initial familiar motivators from E&EE, then explore matrices in depth and then return to examine the broader range of "new to you" applications for E&EE

How is the Package designed to help you learn?



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We have 10 timetabled hours together
All material will be available in Moodle.

What is the purpose of the lectures given that the material is all available? ***Pace, Emphasis and interactivity.*** Most of us can only properly absorb new technical material at a certain rate; my main roles are to set the pace and to dynamically interact with you



Lab. sessions to support the single "Matrices Laboratory" are designed to give you hands on experience

The Overall Point of Matrices?



Physics, data
and signals



Design

$$\begin{pmatrix} \vdots & \dots & \vdots \\ & \ddots & \\ \vdots & \dots & \vdots \end{pmatrix}$$

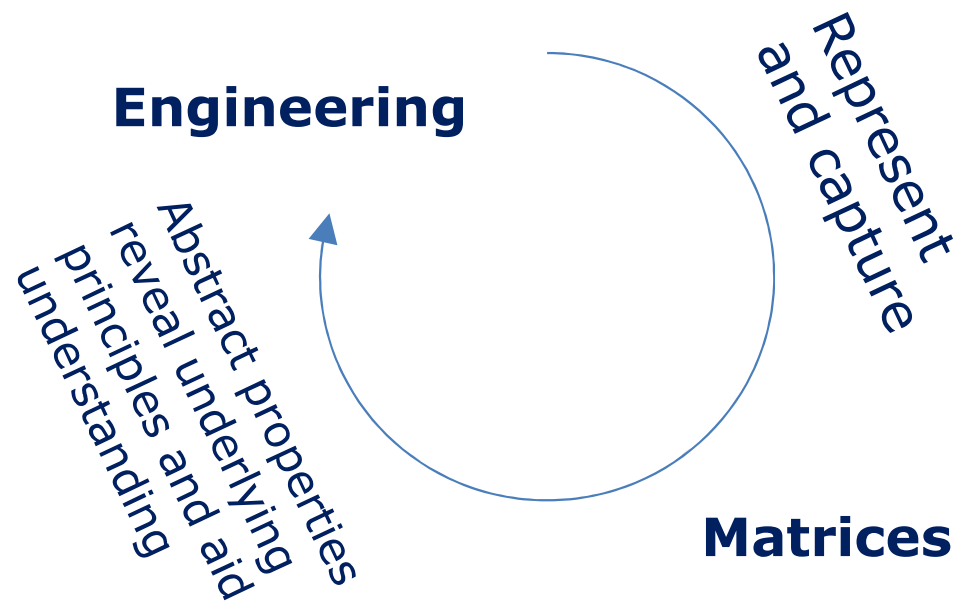
Matrices



Analysis,
Modelling and
Simulation

Matrices and the techniques we use to manipulate them are ***convenient tools*** for the representation and analysis of sets of data and physical equations

**There is a valuable
relationship to
understand**





What is a Matrix (plural Matrices)?

What is a Matrix?

Matrices are "just" a compact and systematic notation for simultaneous equations



Example: *2 equations in 3 unknowns*

$$\begin{array}{rcl} 2 & = & 3x + 9y - 1z \\ 1 & = & 4x + 8y + 2z \end{array} \quad \Rightarrow \quad \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 & 9 & -1 \\ 4 & 8 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

where we understand that we go back from the matrices to the simultaneous equations by "combining row and column pairs", multiplying and adding corresponding terms

$$\begin{pmatrix} 3 & 9 & -1 \\ 4 & 8 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} \text{ this row, column pair gives } 3x + 9y - 1z \text{ which is given to be } 2$$

$$\begin{pmatrix} 3 & 9 & -1 \\ 4 & 8 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} \text{ this row, column pair gives } 4x + 8y + 2z \text{ which is given to be } 1$$

What is a Matrix?



This doesn't look that much more compact!

So now we can introduce a more compressed notation

Let us understand that any symbol written underlined is *representing* a 1 column matrix and any symbol written double underlined is *representing* an n column by m row matrix

Example: 2 equations in 3 unknowns

$$\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 & 9 & -1 \\ 4 & 8 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad \Rightarrow \quad \underline{a} = \underline{\underline{b}} \underline{c}$$

Unfortunately, there is no standardisation on how to denote matrices!

What is a Matrix?



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$$\underline{a} = \underline{\underline{b}} \underline{c}$$

or

$$\bar{a} = \bar{\bar{b}} \bar{c}$$

or

$$\mathbf{a} = \mathbf{b} \mathbf{c}$$

From now on, we will use **bold** to denote any matrices which is the most common these days

Now we have introduced matrices - at least as a compact notation for simultaneous equations - was it worth the effort ?



Why bother with matrices?

(We are engineers not mathematicians)

Why use Matrix Notation?

1) To manage complexity and scale

100's of millions of equations in
100's of millions of unknowns to
capture realistic engineering
challenges

$$a = b c$$

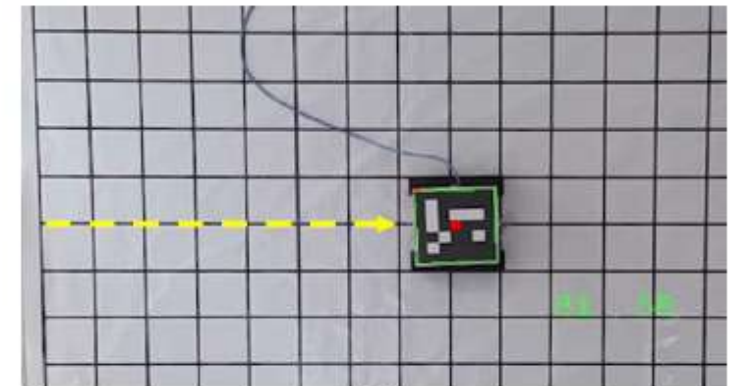
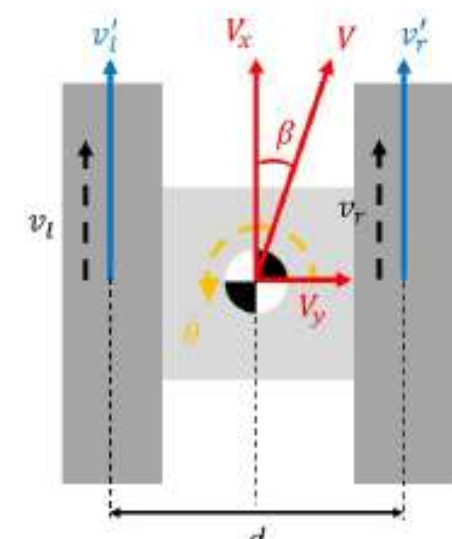
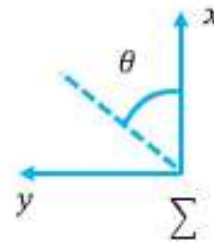
Nice notation at least!

2021 8th International Conference on Computer and Communication Engineering (ICCCE)

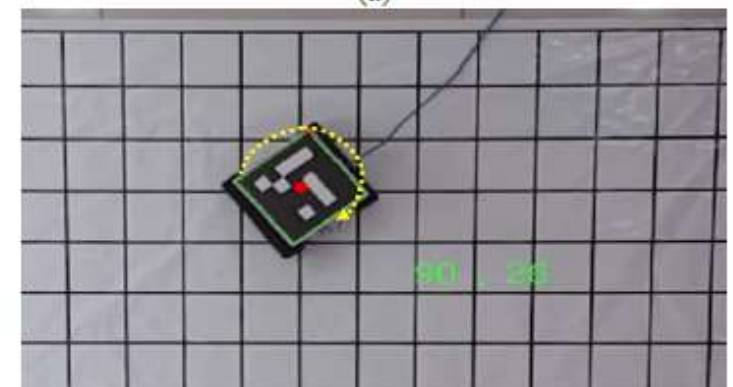
Design of Hardware-in-the-loop Simulation Approach for Slip-Compensated Odometry Tracked Mobile Robot Platform

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(a)



(b)

Fig. 6. (a) Straight Maneuver (b) Turning Maneuver

Why use Matrix Notation?



2) To be systematic ...

We don't solve 100's of millions of equations in 100's of millions of unknowns in the 1st year way - i.e. by hand, *spotting that if we multiply this equation by .. and add it to that equation, then ...* - nightmarish - we adapt the problem,

for computers

Example: Note how the core row-column combining operation is so easy to code

$$a = b c$$

```
float b[9][9],a[9],c[9];
```

```
// Give b and c their known values and set all elements of a to 0
```

```
for(int i=0;i<9;i++){  
    for(int j=0;j<9;j++) a[i]+=b[i][j]*c[j];  
}
```

$$\begin{pmatrix} a_0 \\ a_1 \\ \dots \end{pmatrix} = \begin{pmatrix} b_{00} & b_{01} & \dots \\ b_{10} & b_{11} & \dots \\ \dots & \dots & \dots \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ \dots \end{pmatrix}$$

Understand now why arrays are so prominent in so many computer languages?

Why use Matrix Notation?



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To answer the obvious question here

No, you are not required to do any computer programming in this module!

But we will talk a lot about "computability and programmability"



**Show me how we might *advantageously*
use matrices for a basic engineering task
that I already know**

Where might matrices be relevant to us?

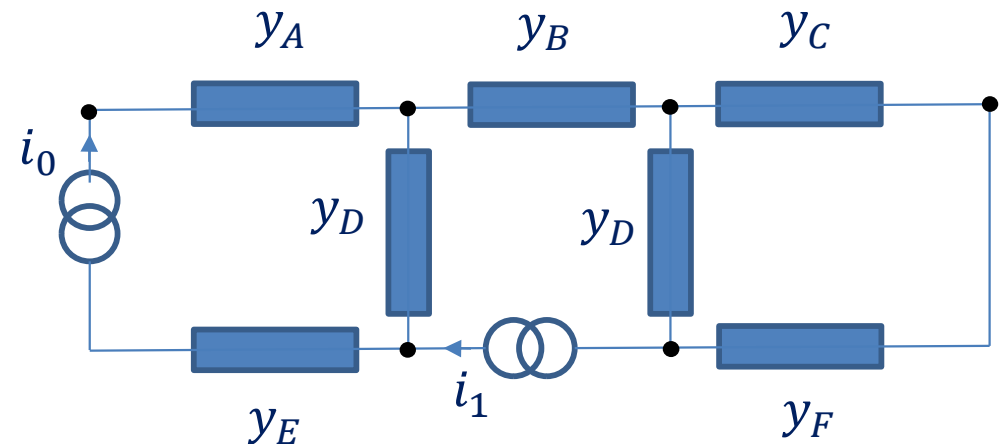


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How matrices can arise and help us manage scale, complexity and computation

Example: *Mesh and Nodal analysis - introduced as a systematic way of solving circuit problems: Here nodal analysis with admittances given for the components and current sources*



The "Nodal Analysis" algorithm:

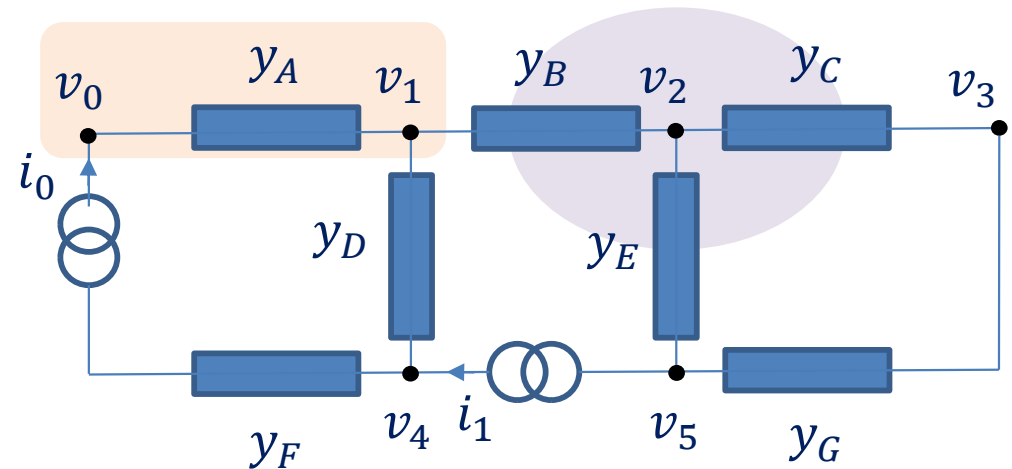
- 1) *Allocate a voltage unknown to each "node" of the circuit*
 - 2) *Express the current through each component in terms of the nodal voltages*
 - 3) *Sum currents at each node (KCL) which leads to simultaneous equations*
- solve for the unknown nodal voltages*

Where might matrices be relevant to us?



2) The current through each component will be their admittance times the voltage difference across them;

e.g. $y_A(v_0 - v_1)$ is the current flowing through component A in the direction from node 0 towards node 1



3) KCL at each node; e.g. the total current flowing out of node 2 is

$$0 = y_B(v_2 - v_1) + y_C(v_2 - v_3) + y_E(v_2 - v_5)$$

Do you spot a pattern that we can exploit to generate the other node's equations?

Maybe rewrite as

$$0 = (y_B + y_C + y_E)v_2 - y_B v_1 - y_C v_3 - y_E v_5$$

Where might matrices be relevant to us?

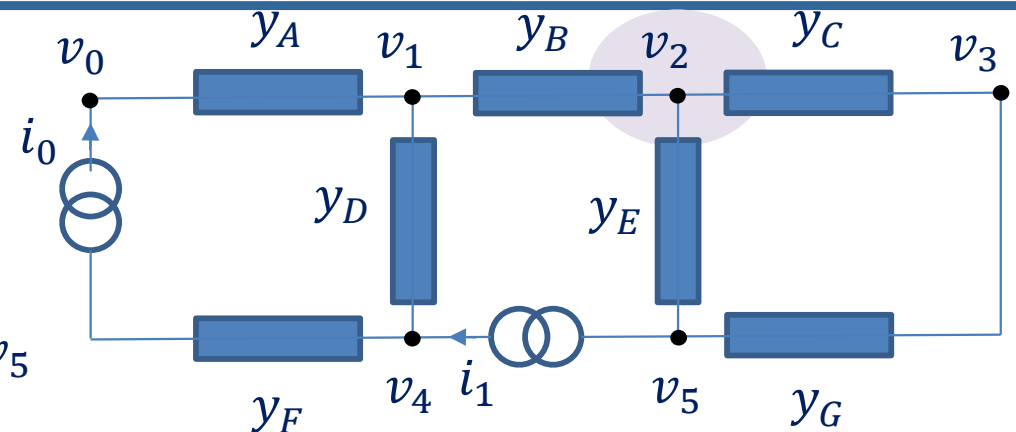


As part of a matrix, this nodal equation,

$$0 = (y_B + y_C + y_E)v_2 - y_B v_1 - y_C v_3 - y_E v_5$$

could be expressed as

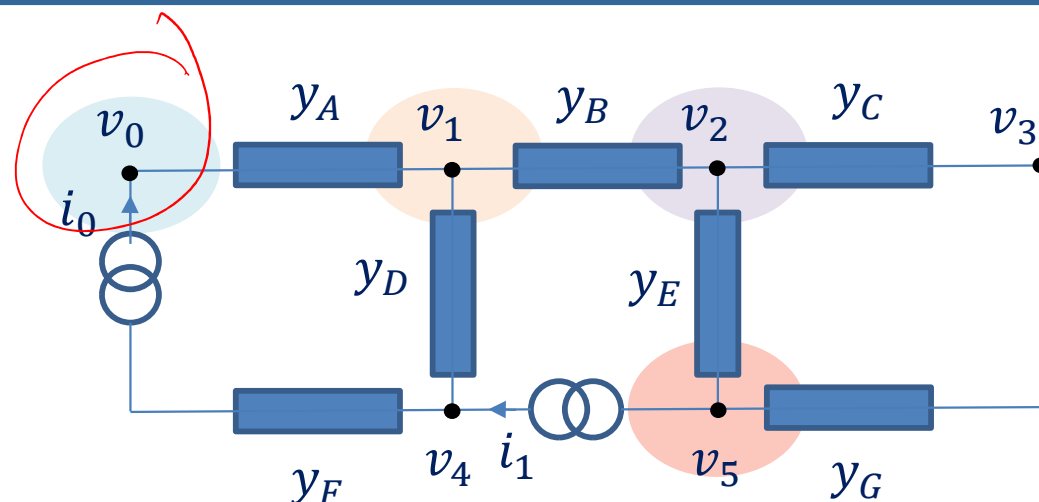
$$\begin{pmatrix} 0 \end{pmatrix} = \begin{pmatrix} 0 & -y_B & (y_B + y_C + y_E) & -y_C & 0 & -y_E \end{pmatrix} \begin{pmatrix} v_0 \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{pmatrix}$$



Although this is 1st year stuff, I am suddenly worried about all sorts of doubts!

But first, the good news: Because we spotted a pattern, filling in the other rows of the matrix is "mechanical" - i.e. I can follow a (programmable!) recipe

Where might matrices be relevant to us?



$$\begin{pmatrix} 0 \\ 0 \\ i_0 \\ -i_1 \end{pmatrix} = \begin{pmatrix} 0 & -y_B & (y_B + y_C + y_E) & -y_C & 0 & -y_E \\ -y_A & (y_A + y_B + y_D) & -y_B & 0 & -y_D & 0 \\ y_A & -y_A & 0 & 0 & 0 & 0 \\ 0 & 0 & -y_E & -y_G & 0 & (y_E + y_G) \end{pmatrix} \begin{pmatrix} v_0 \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{pmatrix}$$

What have we gained by using matrices ?

Maybe if I explain those doubts from the previous slide it might explain

Where might matrices be relevant to us?

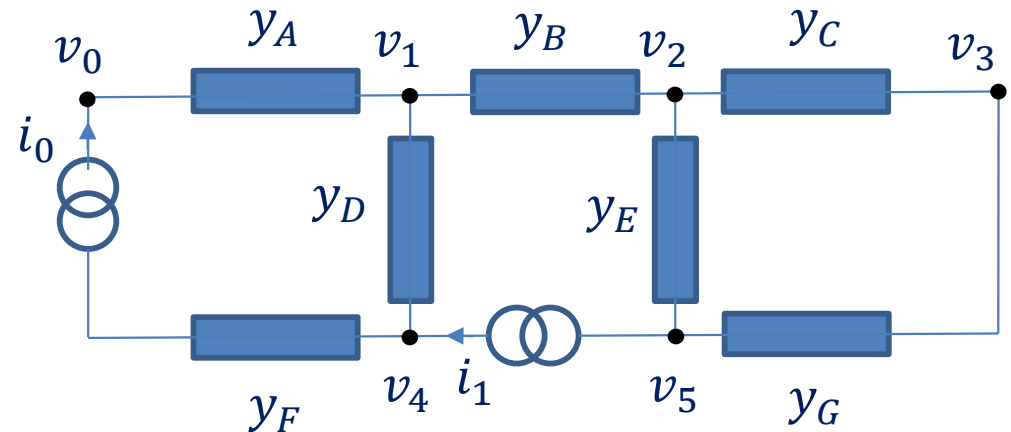


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Maybe if I explain those doubts from the previous slide it might explain

1) I am a human being and make mistakes with voluminous algebra!



2) Did it matter in which order I numbered the voltages?

3) Did it matter in which order I wrote the nodal equations down in the matrix?

4) Did I need to introduce all those voltages - maybe I needed some more?

5) Actually, does the circuit actually have a solution - maybe I have set a physically impossible example - would be embarrassing here!

Where might matrices be relevant to us?



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1) I am a human being and make mistakes with voluminous algebra!

Matrices are very systematic and capture equations in a manner that very often reveals patterns that we can note and use to validate our own processing

2) Did it matter in which order I numbered the voltages?

3) Did it matter in which order I wrote the nodal equations down in the matrix ?

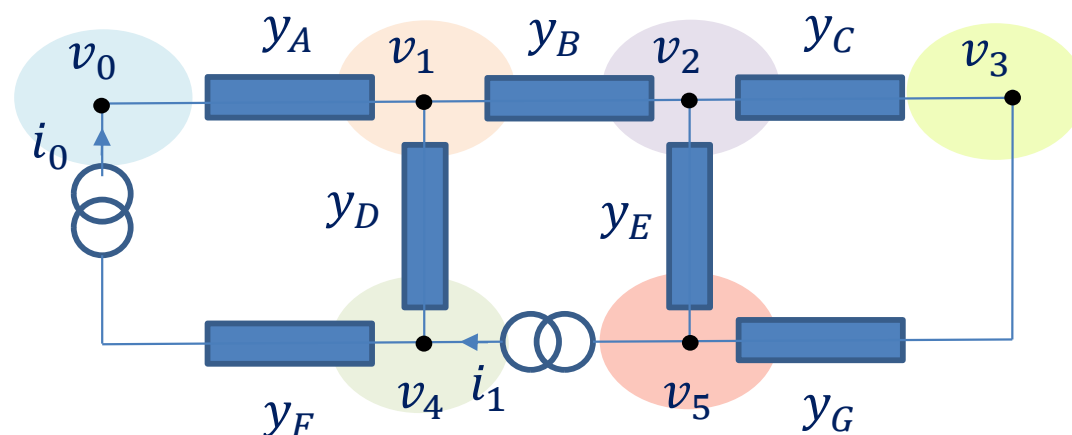
No of course not - the physics here obviously doesn't care about our administrative choices.

But - IF we do make a point of sticking to certain "patterns of doing things" such as the order we write down the nodal equations, then we see even more helpful patterns - reinforcing the 1st point above

Where might matrices be relevant to us?



Recommendation: Process the nodal equations in the order that their voltages are numbered and always apply KCL as total current leaving the node



$$\begin{pmatrix} 0 \\ 0 \\ i_0 \\ 0 \\ i_1 - i_0 \\ -i_1 \end{pmatrix} = \begin{pmatrix} (y_A) & -y_A & 0 & 0 & 0 & 0 \\ -y_A & (y_A + y_B + y_D) & -y_B & 0 & -y_D & 0 \\ 0 & -y_B & (y_B + y_C + y_E) & -y_C & 0 & -y_E \\ 0 & 0 & -y_C & (y_C + y_G) & 0 & -y_G \\ 0 & -y_D & 0 & 0 & (y_D) & 0 \\ 0 & 0 & -y_E & -y_G & 0 & (y_E + y_G) \end{pmatrix} \begin{pmatrix} v_0 \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{pmatrix}$$

What patterns do we see?

Where might matrices be relevant to us ?



$$\begin{pmatrix} 0 \\ 0 \\ i_0 \\ 0 \\ i_1 - i_0 \\ -i_1 \end{pmatrix} = \begin{pmatrix} (y_A) & -y_A & 0 & 0 & 0 & 0 \\ -y_A & (y_A + y_B + y_D) & -y_B & 0 & -y_D & 0 \\ 0 & -y_B & (y_B + y_C + y_E) & -y_C & 0 & -y_E \\ 0 & 0 & -y_C & (y_C + y_G) & 0 & -y_G \\ 0 & -y_D & 0 & 0 & (y_D) & 0 \\ 0 & 0 & -y_E & -y_G & 0 & (y_E + y_G) \end{pmatrix} \begin{pmatrix} v_0 \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{pmatrix}$$

- Generally in matrix notation we have $\mathbf{I}^{N \times 1} = \mathbf{Y}^{N \times N} \mathbf{V}^{1 \times N}$ for nodal analysis and $\mathbf{V}^{N \times 1} = \mathbf{Z}^{N \times N} \mathbf{I}^{1 \times N}$ for mesh analysis
- Diagonal elements are "positive sums" of components connected to the nodes
- Non diagonal elements are all "negative" component values
- Symmetry about the diagonal, e.g.

$$\begin{pmatrix} (y_A) & -y_A & 0 & 0 & 0 & 0 \\ -y_A & (y_A + y_B + y_D) & -y_B & 0 & -y_D & 0 \\ 0 & -y_B & (y_B + y_C + y_E) & -y_C & 0 & -y_E \\ 0 & 0 & -y_C & (y_C + y_G) & 0 & -y_G \\ 0 & -y_D & 0 & 0 & (y_D) & 0 \\ 0 & 0 & -y_E & -y_G & 0 & (y_E + y_G) \end{pmatrix}$$

Where might matrices be relevant to us?



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Patterns are good!

It could be argued that one of the main reasons for using matrices is that it facilitates pattern spotting

Pattern checking - stop us making typos and make computer programming more straightforward

Now to get a bit more subtle, my other doubts

4) Did I need to introduce all those voltages - maybe I needed some more?

5) Actually, does the circuit actually have a solution - maybe I have set a physically impossible example - would be embarrassing here!

Where might matrices be relevant to us?



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These two questions are both related - basically, do we have enough information available to "completely" solve the problem?

Sometimes as engineers, we know this, but as things get more complicated, we can not be sure;

The whole mathematical purpose of matrices is to abstract away from specific engineering problems so that we can return later to make great statements like ...

If an engineering problem can be expressed in matrix form such as

$$\mathbf{i} = \mathbf{Y} \mathbf{v}$$

(which is what we have above)

Then *if* the matrix $\mathbf{Y}$ is symmetric then it is assured that ...



Summary - the reason for us engineers to care about matrices more generally?

Once we can express *our* problems and engineering questions in terms of matrices, then we can draw on a whole repertoire of known properties of matrices to progress

Questions might be of the form:

Is it possible to find a set of inputs to this circuit to give a desired set of outputs?

Is my control loop, with many inputs, outputs and feedback branches, always stable?

What is the most efficient way of compressing the data in images with the least impact on image quality?

Is it possible to rotate a robot arm from one orientation to another - how?



Show me how I might *advantageously* use matrices for a basic engineering task that I already know

Mesh and Nodal analysis - the basis of all circuit simulators such as Spice and PLECS

"Pattern spotting" leads to systematic rules for constructing the overall "maths problem" - hence programmable

"Scalable", i.e. the rules can just be used repeatedly as we ever increase the "size" of the problem (e.g. circuit)

still $a = b \cdot c$

100-by-100

100M-by-100M



$$\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 & 9 & -1 \\ 4 & 8 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

- We say that $\begin{pmatrix} 3 & 9 & -1 \\ 4 & 8 & 2 \end{pmatrix}$ is a 2 (row) by 3 (column) matrix, each entry of which is called an *element* of the matrix. Often we might indicate a particular element in the matrix **a** by subscripts, a_{ij} , i.e. the element in row i, column j. One might explicitly show the size of a matrix using superscripts, i.e. $\mathbf{a}^{(7 \times 3)}$, for emphasis
- For the special case of an n (row) by 1 (column) matrix, e.g. $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ we often refer to it as a "column" or "vector"
- Matrices can generally be N by M with no constraints on N and M except >0
- Matrices can have complex number *elements*

Some basic Matrix operations and definitions



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The operation of combining matrices we introduced through the example like

$$\begin{pmatrix} 3 & 9 & -1 \\ 4 & 8 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 8 \\ -9 \end{pmatrix} = \begin{pmatrix} 3 * 1 + 9 * 8 - 1 * (-9) \\ 4 * 1 + 8 * 8 + 2 * (-9) \end{pmatrix}$$

is called *matrix multiplication* or a *matrix product* and generalises as

$$\begin{pmatrix} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{pmatrix} \begin{pmatrix} * \\ * \\ * \\ * \end{pmatrix} = \begin{pmatrix} * & * \\ * & * \\ * & * \\ * & * \end{pmatrix}$$

$$\mathbf{b}^{(4 \times 3)} \mathbf{c}^{(3 \times 2)} = \mathbf{a}^{(4 \times 2)}$$

where each element of the *output* matrix is obtained by a row-column combination of the two matrices being multiplied

- Combining the i^{th} row of the first matrix with the j^{th} column of the second matrix gives the i,j element of the output matrix
- Note that for the "row-column combine" to make sense, the number of columns of the first matrix *must* be the same as the number of rows of the second



Example: Evaluate the product of the matrices **A** and **B** given by

$$\mathbf{A} = \begin{pmatrix} 3 & 9 & -1 \\ 4 & 8 & 2 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 2 & 5 \end{pmatrix}$$

Solution:



Example: Evaluate the product of the matrices **A** and **B** given by

$$\mathbf{A} = \begin{pmatrix} 3 & 9 & -1 \\ 4 & 8 & 2 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 2 & 5 \end{pmatrix}$$

Solution: The product matrix must have the same number of rows as **A** and the same number of columns as **B**, so is 2 by 3

$$\begin{pmatrix} 3 & 9 & -1 \\ 4 & 8 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 2 & 5 \end{pmatrix}$$

$$= \begin{pmatrix} 3 * 1 + 9 * 0 - 1 * 1 & 3 * 0 + 9 * 2 - 1 * 2 & 3 * 1 + 9 * 0 - 1 * 5 \\ 4 * 1 + 8 * 0 + 2 * 1 & 4 * 0 + 8 * 2 + 2 * 2 & 4 * 1 + 8 * 0 + 2 * 5 \end{pmatrix}$$

$$= \begin{pmatrix} 2 & 16 & -2 \\ 6 & 20 & 14 \end{pmatrix}$$

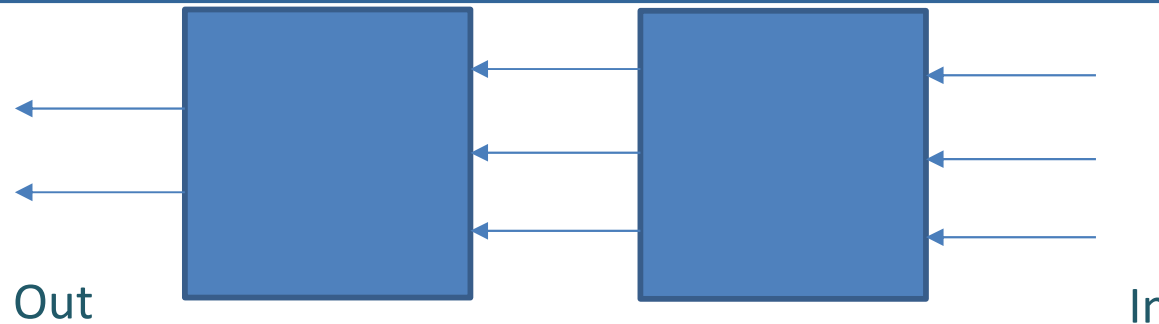
Great - but what does matrix multiplication mean to engineers - what's our picture?

Some basic Matrix operations and definitions



"Circuit/System" A has
3 "inputs" and 2 "outputs"

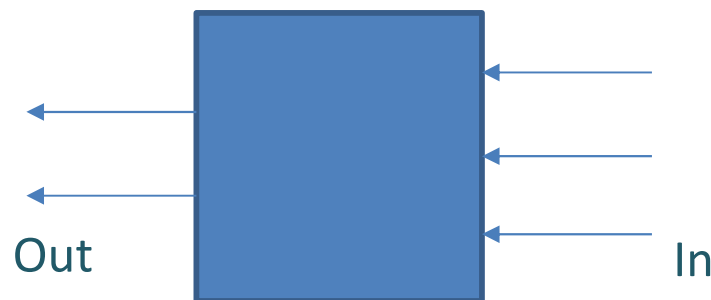
"Circuit/System" B has
3 "inputs" and 3 "outputs"



*Fancy name for a system
that can be represented
like this, "LTI", linear,
time invariant - think on
this!*

$$A = \begin{pmatrix} 3 & 9 & -1 \\ 4 & 8 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 2 & 5 \end{pmatrix}$$

The cascade of A following B behaves like the single "circuit/system" AB



*Matrix multiplication is the mathematics of
"cascades" - sounds useful?*

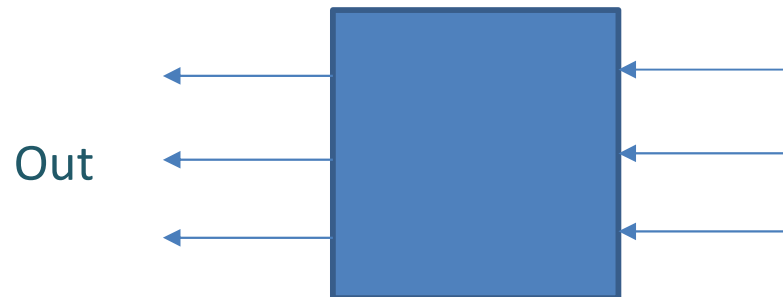
$$AB = \begin{pmatrix} 2 & 16 & -2 \\ 6 & 20 & 14 \end{pmatrix}$$



Example: Doesn't the circuit picture not make it obvious why the product of the matrices BA is meaningless (or impossible)

Solution:

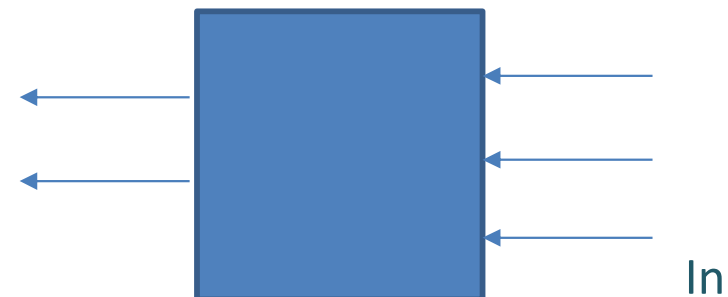
"Circuit/System" B has
3 "inputs" and 3 "outputs"



$$B = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 2 & 5 \end{pmatrix}$$

Connection mismatch!

"Circuit/System" A has
3 "inputs" and 2 "outputs"



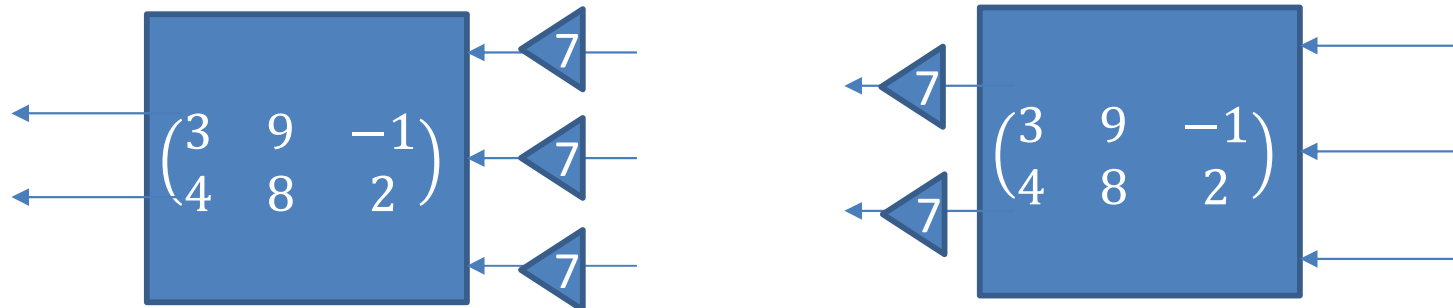
$$A = \begin{pmatrix} 3 & 9 & -1 \\ 4 & 8 & 2 \end{pmatrix}$$



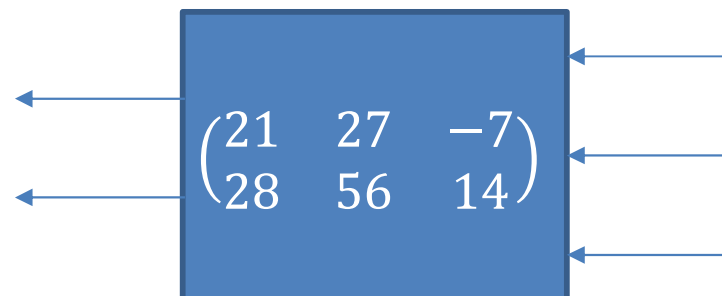
We can similarly use this way of thinking for justifying the use of the *scalar product*

The scalar product is obtained by scaling each element of a matrix by the same amount, e.g.

$$7 * \begin{pmatrix} 3 & 9 & -1 \\ 4 & 8 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 9 & -1 \\ 4 & 8 & 2 \end{pmatrix} * 7 = \begin{pmatrix} 21 & 27 & -7 \\ 28 & 56 & 14 \end{pmatrix}$$



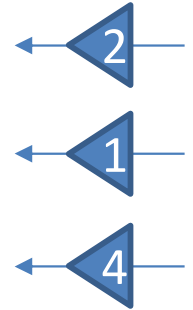
Both configurations behave like the single "circuit/system"



Scalar multiplication is the mathematics of "amplification"



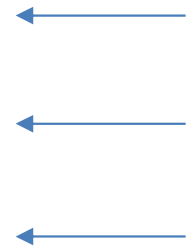
Example: What is the matrix representation of the following circuit?



i.e. each output of the "system" is one of the inputs amplified

Solution:

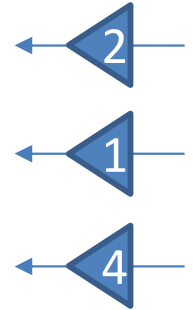
Example: What is the matrix representation of the following circuit?



i.e. each output of the "system" is connected to one of the inputs

Solution:

Example: What is the matrix representation of the following circuit?

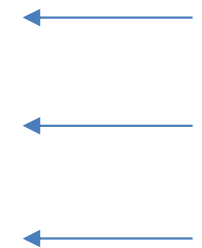


i.e. each output of the "system" is one of the inputs amplified

Solution:

$$\begin{pmatrix} v_{o0} \\ v_{o1} \\ v_{o2} \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix} \begin{pmatrix} v_{i0} \\ v_{i1} \\ v_{i2} \end{pmatrix}$$

Example: What is the matrix representation of the following circuit?



i.e. each output of the "system" is connected to one of the inputs

Solution:

$$\begin{pmatrix} v_{o0} \\ v_{o1} \\ v_{o2} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} v_{i0} \\ v_{i1} \\ v_{i2} \end{pmatrix}$$



The matrix of the last example, in engineering speak "outputs=inputs" or a simple "multi-line connection" is actually a rather useful concept here; it has the same role as multiplying by "1" in simple arithmetic.

Mathematically we call such matrices **Identity Matrices**, noting that the number of rows is the same as the number of columns. It is usually given the symbol **I** or **1**

Example: Give a 4 by 4 identity matrix

Solution:

$$I = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$AI = IA = A$$
$$3 \times 1 = 1 \times 3 = 3$$

*Note that the whole point is that **A I = I A = A** for any suitably sized **A***



To complete the set, we also define **matrix addition and subtraction** as an *elementwise* sum and difference; the matrices must be of the same size

Example: Evaluate the sum of the matrices A and B given by

$$A = \begin{pmatrix} 3 & 9 & -1 \\ 4 & 8 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 & 3 \\ 1 & 2 & 5 \end{pmatrix}$$

Solution:

Example: Evaluate the difference of the matrices A and B given by

$$A = \begin{pmatrix} 3 & 9 & -1 \\ 4 & 8 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 & 3 \\ 1 & 2 & 5 \end{pmatrix}$$

Solution:



To complete the set, we also define **matrix addition and subtraction** as an *elementwise* sum and difference; the matrices must be of the same size

Example: Evaluate the sum of the matrices A and B given by

$$A = \begin{pmatrix} 3 & 9 & -1 \\ 4 & 8 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 & 3 \\ 1 & 2 & 5 \end{pmatrix}$$

Solution:

$$A + B = \begin{pmatrix} 3+1 & 9+0 & -1+3 \\ 4+1 & 8+2 & 2+5 \end{pmatrix} = \begin{pmatrix} 4 & 9 & 2 \\ 5 & 10 & 7 \end{pmatrix}$$

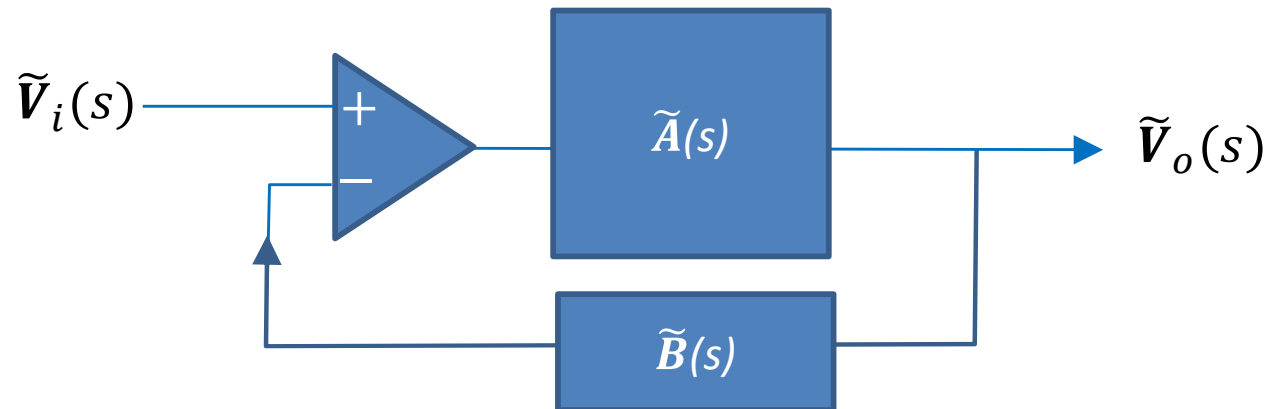
Example: Evaluate the difference of the matrices A and B given by

$$A = \begin{pmatrix} 3 & 9 & -1 \\ 4 & 8 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 & 3 \\ 1 & 2 & 5 \end{pmatrix}$$

Solution:

$$A - B = \begin{pmatrix} 3-1 & 9-0 & -1-3 \\ 4-1 & 8-2 & 2-5 \end{pmatrix} = \begin{pmatrix} 2 & 9 & -4 \\ 3 & 6 & -3 \end{pmatrix}$$

Example: Consider a multi-input, multi-output control loop; this illustrates addition and subtraction for us



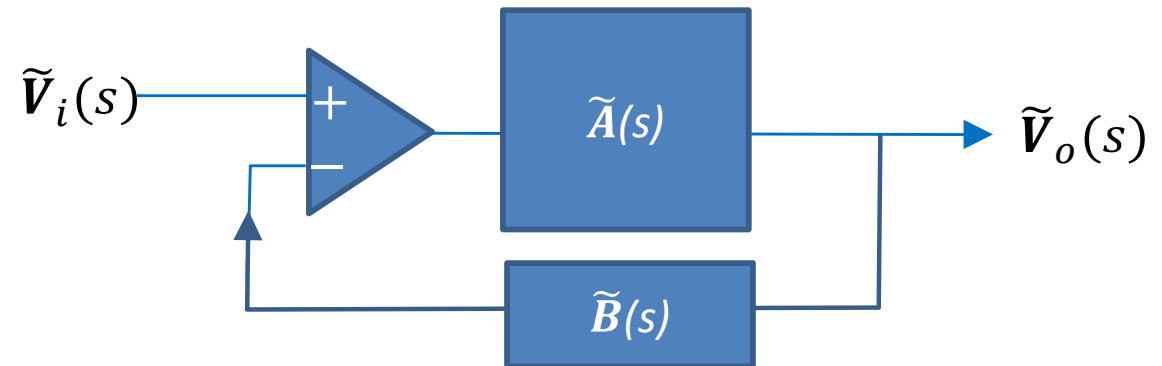
Note that $\tilde{A}$ and $\tilde{B}$ are matrices, the elements of which are Laplace Transforms

That's OK, LTs are linear and who said matrices must only contain constant numbers - could be functions - that's both allowed and massively increases the usefulness

For example ...

... if we say

$$A = \begin{pmatrix} 1 & \frac{1}{s+2} \\ \frac{1}{s+3} & 2 \end{pmatrix} \quad B = \begin{pmatrix} s & 2 \\ -9 & \frac{2}{s} \end{pmatrix}$$



We could consider - the usual proof, but with multiple inputs/outputs - hence convenient to use matrices. I clearly assume 2 inputs and 2 outputs here

$$V_o = A (V_i - B V_o)$$

$$V_o - A B V_o = A V_i$$

$$(I + A B) V_o = A V_i$$

Note the use of a 2 by 2 identity matrix - to act like "1"

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & \frac{1}{s+2} \\ \frac{1}{s+3} & 2 \end{pmatrix} \begin{pmatrix} s & 2 \\ -9 & \frac{2}{s} \end{pmatrix} V_o = \begin{pmatrix} 1 & \frac{1}{s+2} \\ \frac{1}{s+3} & 2 \end{pmatrix} V_i$$

Matrix product

Matrix addition

Some basic Matrix operations and definitions



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$$(I + AB) V_o = A V_i$$

$$\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & \frac{1}{s+2} \\ \frac{1}{s+3} & 2 \end{pmatrix} \begin{pmatrix} s & 2 \\ -9 & \frac{2}{s} \end{pmatrix} \right) V_o = \begin{pmatrix} 1 & \frac{1}{s+2} \\ \frac{1}{s+3} & 2 \end{pmatrix} V_i$$

$$\left(\begin{pmatrix} 1 + s - \frac{9}{s+2} & 0 + 2 + \frac{2}{s(s+2)} \\ 0 + \frac{s}{s+3} - 18 & 1 + \frac{2}{s+3} + \frac{4}{s} \end{pmatrix} \right) V_o = \begin{pmatrix} 1 & \frac{1}{s+2} \\ \frac{1}{s+3} & 2 \end{pmatrix} V_i$$

So the point of this example was:

- 1) To show in context use of matrix multiplication
- 2) To show in context use of matrix addition
- 3) To show in context use of an identity matrix
- 4) A gentle illustration of how matrix elements need not just be numbers

Do we feel we have "solved" the problem?, Surely we want $V_o = \dots$



Do we feel we have "solved" the problem?, Surely we want $V_o = \dots$

$$(I + AB) V_o = A V_i$$

$$\begin{pmatrix} 1 + s - \frac{9}{s+2} & 0 + 2 + \frac{2}{s(s+2)} \\ 0 + \frac{s}{s+3} - 18 & 1 + \frac{2}{s+3} + \frac{4}{s} \end{pmatrix} V_o = \begin{pmatrix} 1 & \frac{1}{s+2} \\ \frac{1}{s+3} & 2 \end{pmatrix} V_i$$

Let's denote the left matrix as $\mathbf{C}$ so that $\mathbf{C} V_o = A V_i$

So hands up if we can now write

$$V_o = \frac{A}{C} V_i$$

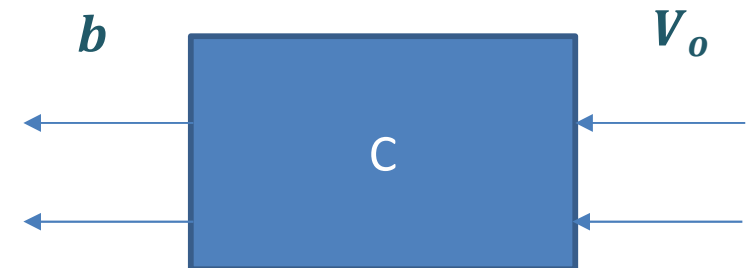
If your hand is up, then either pretend your just stretching, or else I would argue if I were you that you just meant it was OK as a "concept"

Matrix division does not exist in the conventional sense!

$$C V_o = A V_i$$

Remembering that A and V_i are known matrices, their product can be evaluated, so we are faced with

$$C V_o = b$$



As engineers we recall the circuit interpretation of matrix multiplication?

So rather than being given the inputs and trying to find the outputs:

Our task expressed by $C V_o = b$ is; given the outputs b what are the inputs V_o

So hands up if we can now write?

$$V_o = C^{-1}b = \underline{C^{-1}A}V_i$$

Well yes we can write it - but ...



Summary: We are often confronted with problems of the form $A\mathbf{x} = \mathbf{b}$ where A is a known matrix and $\mathbf{b}$ a known column and we have to find column $\mathbf{x}$

In fact, this is probably the most commonly occurring task for computational scientists and engineers

Later we will remind ourselves of the 1st year concept of the inverse matrix which claims to do this; given a matrix A we can find another matrix, its *inverse and denoted by* A^{-1} that "undoes" A , by which we mean $A^{-1} A = A A^{-1} = I$, hence $\mathbf{x} = A^{-1}\mathbf{y}$

Now the real world!

- 1) The matrix inverse A^{-1} may not actually exist!
- 2) Numerically, use of the inverse is pathologically doomed to failure!
- 3) The CPU time (and memory) needed to evaluate $A^{-1}\mathbf{y}$ is appallingly bad

We can "solve" $A\mathbf{x} = \mathbf{y}$ but we need to be much smarter - hence this module!



**What is the recommended way of solving
 $Ax = b$ then if it is so common?**

- No - we aren't equipped with enough tools yet - next week!

Back to some basic definitions:

The *transpose* of a matrix is defined to be a "reflection along the diagonal line" and denoted by the superscript T

Example: Give the transposes of the matrices

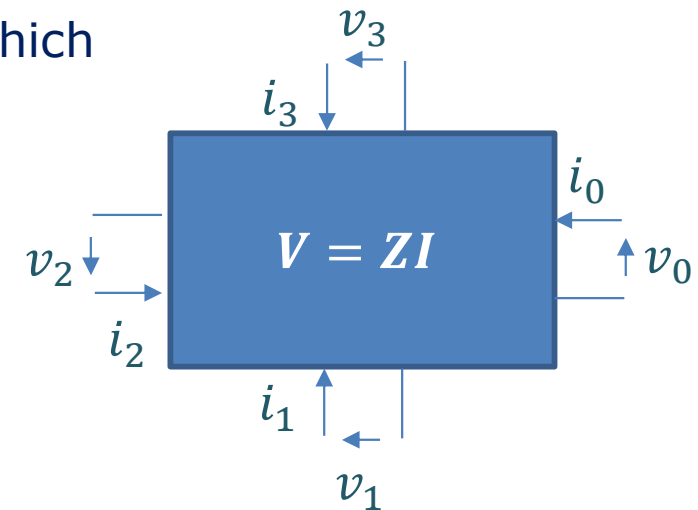
$$\mathbf{A} = \begin{pmatrix} 3 & 9 & -1 \\ 4 & 8 & 2 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 2 & -4 \\ 5 & 1 \end{pmatrix} \quad \mathbf{C} = (1 \quad 2 \quad 3)$$

Solution:

$$\mathbf{A}^T = \begin{pmatrix} 3 & 4 \\ 9 & 8 \\ -1 & 2 \end{pmatrix} \quad \mathbf{B}^T = \begin{pmatrix} 2 & 5 \\ -4 & 1 \end{pmatrix} \quad \mathbf{C}^T = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

Let's look at an engineering example where the concept of the transpose arises

Example: A circuit has a number of ports, each of which has a voltage and current defined. The circuit behaviour can be represented as a matrix impedance, $\mathbf{V} = \mathbf{Z}\mathbf{I}$, where the elements of matrix $\mathbf{Z}$ are real (i.e. resistances). Give a simple expression for the total power absorbed by the circuit



Solution: We introduce the columns

$$\mathbf{V} = \begin{pmatrix} v_0 \\ v_1 \\ v_2 \\ v_3 \end{pmatrix} \quad \mathbf{I} = \begin{pmatrix} i_0 \\ i_1 \\ i_2 \\ i_3 \end{pmatrix} \quad \text{The total power is } P = \sum_{j=0}^{N-1} v_j i_j \text{ which could be expressed as}$$

$$P = (v_0 \quad v_1 \quad v_2 \quad v_3) \begin{pmatrix} i_0 \\ i_1 \\ i_2 \\ i_3 \end{pmatrix} = \mathbf{V}^T \mathbf{I}$$

To pursue this further, we'd need to transpose the product $\mathbf{V} = \mathbf{Z}\mathbf{I}$



The transpose of a matrix product can be expressed as

$$(A B C D)^T = D^T C^T B^T A^T$$

Returning to the example, $V^T = I^T Z^T$

$$P = (v_0 \quad v_1 \quad v_2 \quad v_3) \begin{pmatrix} i_0 \\ i_1 \\ i_2 \\ i_3 \end{pmatrix} = V^T I = I^T Z^T I$$

Note how we have evaluated a single number as an answer from multi-element matrices. This is typical of where one encounters the transpose.

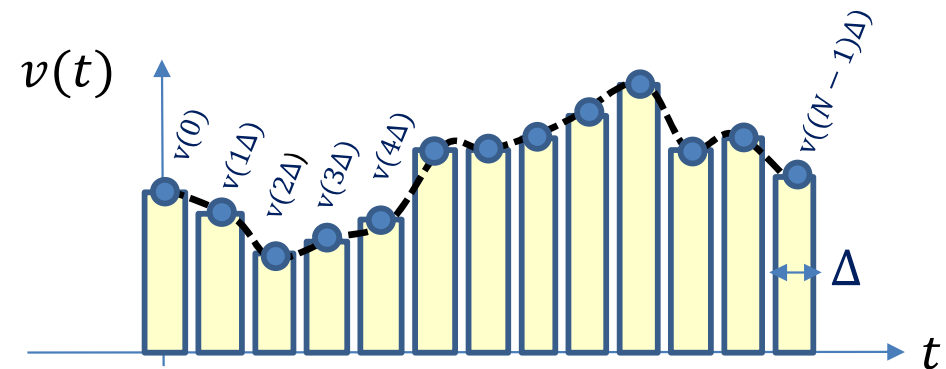
A little challenge! I promised that abstract matrix theory can illuminate engineering. Consider repeating the above analysis in a slightly different way, surely it is just as valid to write

$$P = I^T V = I^T Z I$$

A different answer? Conclusion ?

Example: In a time sampled system measurements of a voltage are made at intervals of Δ seconds. Express in matrix notation an estimation of the RMS voltage

$$V_{RMS}^2 = \frac{1}{T} \int_0^T v^2 dt \sim \frac{1}{(N-1)\Delta} \sum_{n=0}^{N-1} \Delta v_n^2$$



Solution: We introduce the column $\mathbf{V} = \begin{pmatrix} v(0\Delta) \\ v(1\Delta) \\ v(2\Delta) \\ \vdots \end{pmatrix}$

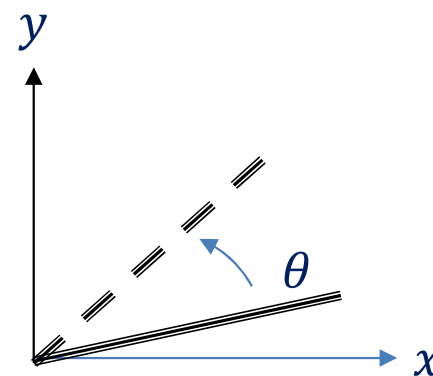
$$V_{RMS}^2 = \frac{1}{(N-1)} \mathbf{V}^T \mathbf{V}$$

Note: For the same reasons of convenience that we use RMS for circuit signals to give an "average" strength for a signal, the quantity $V^T V$ is understood to give a useful scalar magnitude of a column of elements; This is an example of what we call the **norm** of a vector/column and given a symbol

$$\|V\|_2^2 = V^T V$$

This leads nicely into another useful application area for matrices for us - robotics

Example: If in a 2D world, the end of a robotic arm which is initially at coordinates x_i, y_i , is rotated θ radians about its origin as shown. Express in matrix notation the coordinates of the end of the arm, x_o, y_o , after rotation

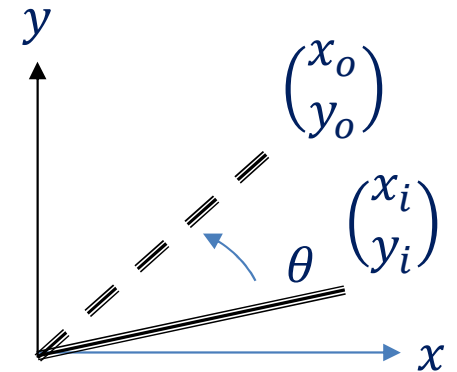


Solution: Introduce a "position vector"

$$\mathbf{r} = \begin{pmatrix} x \\ y \end{pmatrix}$$

A little test of your geometry will show that

$$\begin{pmatrix} x_o \\ y_o \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x_i \\ y_i \end{pmatrix}$$



A little validation check. Surely the length of the arm doesn't change ?

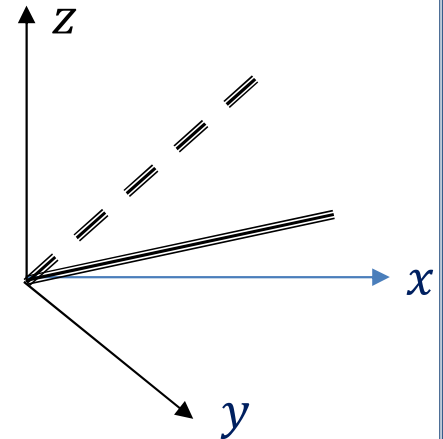
In matrix notation we expect $\|\mathbf{r}_o\|_2 = \|\mathbf{r}_i\|_2$ is this the case ? You check!

Later on we shall make much use of such "rotation" matrices - suitably abstracted of course to gain maximum benefit.

Some basic Matrix operations and definitions



Example: In the previous example, but now the world is 3D,
Hint: don't attempt to do 3D geometry - build it up from
a cascade of 2D rotations



$$\mathbf{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

If the rotation were constrained to lie in the x-y plane then the z coordinate doesn't change, so

$$\begin{pmatrix} x_o \\ y_o \\ z_o \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_i \\ y_i \\ z_i \end{pmatrix}$$

Note: We can tell it's a rotation *about* the z axis as $z_o = z_i$

Follow this by a rotation in just the y-z plane, then one is just in the z-x plane

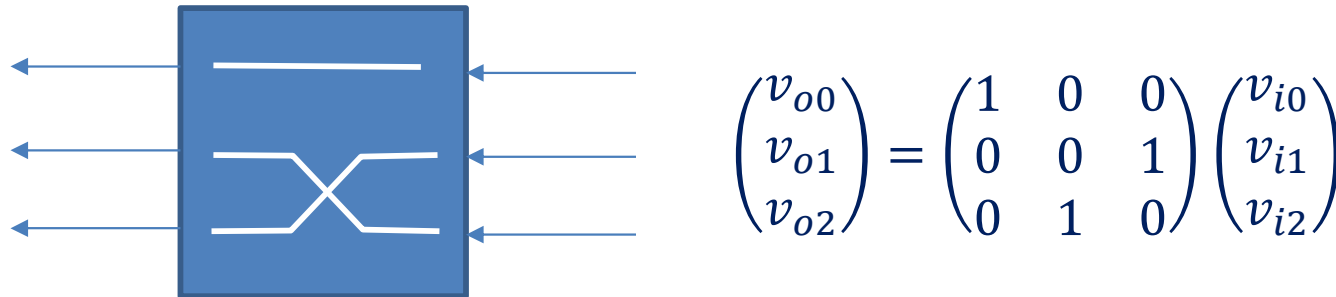
$$\begin{pmatrix} x_o \\ y_o \\ z_o \end{pmatrix} = \begin{pmatrix} \cos \varphi & 0 & -\sin \varphi \\ 0 & 1 & 0 \\ \sin \varphi & 0 & \cos \varphi \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \gamma & -\sin \gamma \\ 0 & \sin \gamma & \cos \gamma \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_i \\ y_i \\ z_i \end{pmatrix}$$

Can we be sure that this can represent *any* repositioning of the end of the arm?

Example: I have a communications network which contains a router/switch. The router has N input channels each of which is connected to a unique output channel. Express the "systems" operation of the router in matrix notation



Solution:



A special case of our general circuit picture of matrix multiplication where each row and column of the matrix contains only a single non-zero element - value 1

Such a matrix performs a *permutation* ? Hence known as *permutation matrix*

Surely such matrices have some helpful properties? How about $\|v_o\|_2 = \|v_i\|_2$?

Last point; engineer uses their understanding to deduce a maths result!

Example: We have undertaken a nodal analysis as we saw choosing a particular ordering of the unknown voltages obtaining $I^{N \times 1} = Y^{N \times N} V^{1 \times N}$. How could we easily obtain the impedance matrix if the ordering of the voltages were changed?

Solution: Isn't the previous example just like a reordering? For example create two different columns from the same voltages

$$V = \begin{pmatrix} v_0 \\ v_1 \\ v_2 \end{pmatrix} \quad V' = \begin{pmatrix} v_2 \\ v_1 \\ v_0 \end{pmatrix} \quad \text{Clearly } V' \text{ is just a reordering of } V \text{ so here } V' = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} V$$

In general there must exist (a lovely phrase!) a permutation matrix P such that a reordering is represented by $V' = P V$ or the other way round $V = Q V'$

We have $I = Y V$ and we want to find Y' such that $I' = Y' V'$

$$Q I' = Y Q V'$$

$$I' = Q^{-1} Y Q V' = [Q^{-1} Y Q] V' \quad \text{hence} \quad Y' = Q^{-1} Y Q$$

Some Practice Problems

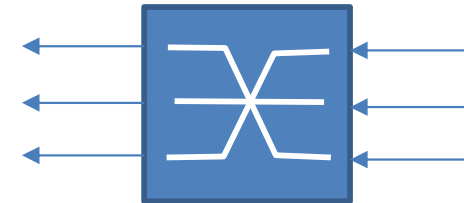


Example: We worried about the use of the inverse earlier - yet have just used it.

Show that the inverse of a (square) permutation matrix P is P^T another permutation matrix and hence "safe to use": i.e. is $P^T P = I$?

Solution: Intuitive - reverse the connections ? Proof by example - dodgy really!

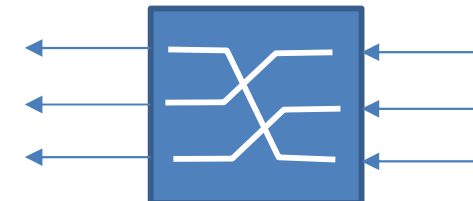
$$V = \begin{pmatrix} v_0 \\ v_1 \\ v_2 \end{pmatrix} \quad V' = \begin{pmatrix} v_2 \\ v_1 \\ v_0 \end{pmatrix} \quad V' = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} V$$



$$P = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad P^T = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad P^T P = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I$$

or

$$V = \begin{pmatrix} v_0 \\ v_1 \\ v_2 \end{pmatrix} \quad V' = \begin{pmatrix} v_2 \\ v_0 \\ v_1 \end{pmatrix} \quad V' = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} V$$



$$P = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad P^T = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad P^T P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I$$

Some Practice Problems



Example: We have an N by M matrix, A that represents some engineering simultaneous equations. Express as a matrix operation keeping the "unknowns" in the same order, but changes the ordering of the equations to get a new matrix A'

Solution: Our very first example was a "two equations in 3 unknowns" set of simultaneous equations. So changing the order we write down the equations means "how to swap rows"

$$\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 & 9 & -1 \\ 4 & 8 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

In general there must exist (that lovely phrase!) a permutation matrix P_R that represents the reordering the rows, so $A' = P_R A$. Let's check on our example

$A = \begin{pmatrix} 3 & 9 & -1 \\ 4 & 8 & 2 \end{pmatrix}$, let's suggest $P_R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and hence

$$A' = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 3 & 9 & -1 \\ 4 & 8 & 2 \end{pmatrix} = \begin{pmatrix} 4 & 8 & 2 \\ 3 & 9 & -1 \end{pmatrix} - \text{i.e. the rows are swapped as desired}$$

Some Practice Problems



Example: We have an N by M matrix, A that represents some engineering simultaneous equations. Express as a matrix operation reordering the "unknowns" but keeping the ordering of the equations the same to get a new matrix A'

Solution: So this is the "dual", asking us for just a column reordering using a permutation matrix, $A' = A P_C$

Couple of worthwhile points:

Convince yourself that the 1st example of this set, the nodal equations is just doing *both* a row and a column reordering with $P_C = P_R^T$ (*which we also know is P_R^{-1}*)

We will find later that although reordering the rows and columns does not affect the "theoretical" "answers" to engineering problems, it does a major impact on the numerical reliability of the results (Think about finite computer precision - maybe only 8 sig. dig.)

Some Practice Problems



Example: We have an N by M matrix, A . Design a matrix M such that MA has the effect of scaling the 2nd row of A by 7

Solution:

$$\text{If } A = \begin{pmatrix} A_{00} & A_{01} & A_{02} & A_{03} & A_{04} \\ A_{10} & A_{11} & A_{12} & A_{13} & A_{14} \\ A_{20} & A_{21} & A_{22} & A_{23} & A_{24} \\ A_{30} & A_{31} & A_{32} & A_{33} & A_{34} \end{pmatrix}$$

then

$$MA = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 7 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} A_{00} & A_{01} & A_{02} & A_{03} & A_{04} \\ A_{10} & A_{11} & A_{12} & A_{13} & A_{14} \\ A_{20} & A_{21} & A_{22} & A_{23} & A_{24} \\ A_{30} & A_{31} & A_{32} & A_{33} & A_{34} \end{pmatrix} = \begin{pmatrix} A_{00} & A_{01} & A_{02} & A_{03} & A_{04} \\ 7A_{10} & 7A_{11} & 7A_{12} & 7A_{13} & 7A_{14} \\ A_{20} & A_{21} & A_{22} & A_{23} & A_{24} \\ A_{30} & A_{31} & A_{32} & A_{33} & A_{34} \end{pmatrix}$$

Example: We have an N by M matrix, A . Design a matrix M such that MA has the effect of adding and subtracting the 2nd and 3rd rows, i.e. if

$$A = \begin{pmatrix} A_{00} & A_{01} & A_{02} & A_{03} & A_{04} \\ A_{10} & A_{11} & A_{12} & A_{13} & A_{14} \\ A_{20} & A_{21} & A_{22} & A_{23} & A_{24} \\ A_{30} & A_{31} & A_{32} & A_{33} & A_{34} \end{pmatrix} \text{ select } M \text{ such that}$$

$$MA = \begin{pmatrix} A_{00} & A_{01} & A_{02} & A_{03} & A_{04} \\ A_{10} + A_{20} & A_{11} + A_{21} & A_{12} + A_{22} & A_{13} + A_{23} & A_{14} + A_{24} \\ A_{10} - A_{20} & A_{11} - A_{21} & A_{12} - A_{22} & A_{13} - A_{23} & A_{14} - A_{24} \\ A_{30} & A_{31} & A_{32} & A_{33} & A_{34} \end{pmatrix}$$

Solution:

$$M = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$



Initially matrices are just a notation for simultaneous equations: *Learning*

Objective - get your head around writing down the matrix form of simultaneous equations

We saw that patterns appear in the matrices if we are logical in the ordering of equations and unknowns; Patterns are good for programs!

We have described and illustrated by example

matrix multiplication - *think circuit cascades*

Scalar multiplication - *think amplifiers*

Identify matrices - *think simple wiring connections*

Transposing - seems to be useful in obtaining single numbers such as
"total power", "length of"

We briefly noted the role of *rotation* matrices in for example robotics

We explored permutation matrices



We noted that the engineering task of finding x when $Ax = b$ for given A and b is not as easy as we thought ($x = A^{-1}b$) - a lot more to do on this as it is crucial! But if I leave you with the best part of the lecture; That phrase

"In general there must exist ..."

by which I am saying that because I have expressed my engineering problem in matrix form, I can quote a myriad of useful matrix theory results that for us might mean

"This control loop must be stable"

"There must be a solution to this set of simultaneous equations"

"This circuit does not absorb any energy"

The skill of engineering mathematics is knowing what abstract mathematics results are out there and which can answer really practical engineering questions"